

Unit 1.4

Rate and Speed

Paper 1: Basic

1. The table shows information collected by Huma about her driving in 2015.

Total distance driven in 2015	19 629 km
Average price paid for petrol	\$2.42 per litre
Average petrol consumption of her car	6.7 litres per 100 km

Calculate the total amount Huma paid for petrol in 2015.

[2]

[N16/I/19(a)]

Paper 1: Intermediate

2. A car travels at an average speed of 82.5 km/h for 3.35 hours.

(a) By rounding these numbers correct to 1 significant figure, find an estimate of the distance travelled by the car.
Show the numbers you use.

[1]

(b) Without doing any further calculation, explain why the actual distance travelled by the car is greater than the answer to part (a).

[1]

[N21/I/3]

3. An aeroplane leaves Adelaide, in Australia, at 09 25 local time.

The distance from Adelaide to Singapore is 5408 km.

The average speed of the aeroplane is 721 km/h.

The aeroplane arrives in Singapore at 15 25 local time.

Find the time difference between Adelaide and Singapore, stating whether the time in Singapore is ahead of or behind the time in Adelaide.

Show your working.

[3]

[N18/I/10]

4. Gina can paint 7 fence panels in 5 hours.

Lim can paint 6 fence panels in 4 hours.

Gina and Lim work together to paint a total of 17 panels.

If they continue to paint at the same rate, how long will it take them to paint the 17 panels?

Give your answer in hours and minutes, to the nearest minute.

[3]

[N15/I/13]

